ORB for Stop Sign Detection

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***Abstract*—Feature matching for object detection is a fundamental concept used in many autonomous vehicles. In this paper, we evaluate one specific algorithm, ORB (Oriented FAST and Rotated Brief), and RANSAC in the context of road sign detection. We acquire a dataset of images and videos of stop signs across the city of Boston to test the algorithm in an uncontrolled environment. We achieved high accuracy for the detection of nearby signs, but struggled to detect signs from a distance. We evaluate the accuracy based on images and videos taken at various camera angles and demonstrate ORB's limitations for scale variation.**

I. Introduction & Motivation

Object detection is important in the context of computer vision problems involving autonomous vehicles and navigation. In urban environments such as Boston, stop signs can be missed by drivers due to congestion, visual clutter, and complex pedestrian intersections. This makes stop-sign detection important for driver assistance systems and autonomous vehicles operating in real-world traffic conditions. Studying ORB-based stop sign detection within Boston's driving conditions offers insight into how classical feature-based object detection methods perform in real-world scenarios. [1] The algorithm previously proposed, ORB (oriented FAST and rotated BRIEF), is advantageous compared to other feature detection algorithms in terms of latency. ORB is cited in many papers and widely used throughout robotics due to its efficiency and it being completely open source, and simple to implement. We chose ORB to implement since it is a classical corner detection method and stop signs have sharp, distinct corners. Similarly, ORB is known to solve rotation challenges, which is relevant in our situation as road signs are often viewed at an angle to moving vehicles or pedestrians instead of head-on. Furthermore, we selected ORB instead of a modern deep learning detector to explore a more computationally simpler and efficient algorithm that matches the scope of our course.

II. Problem Statement

Object detection can be challenging in urban areas and situations where lighting is inconsistent, the camera is at an angle from the detected object, or where the camera is far from the detected object. In this paper we evaluate ORB's effectiveness at detecting stop signs in urban environments. We input a reference stop sign template and a dataset of input images and videos. We modify and implement ORB to determine true or false if a stop sign is present and evaluate the accuracy.

III. Proposed Solution

Our proposed system uses ORB for feature detection and RANSAC for geometric verification to determine the presence of a stop sign.

A. ORB + RANSAC

Oriented FAST and rotated BRIEF (ORB) adds rotational orientation to previously existing FAST and BRIEF techniques. ORB implements FAST to detect keypoints and BRIEF to describe them. It improves on both of these previously existing techniques to add rotation invariance which allows us to detect stop signs captured from more viewpoints and angles. RANSAC filters out false matches and generates a homography to make sure there is geometric consistency between the reference images and the test images.

B. FAST and oFAST: Keypoint Detection

For each image we apply oFAST to find corner points. FAST examines a circle of 16 pixels around a center pixel. If at least 9 contiguous pixels are significantly brighter or darker than the center it labels this as the keypoint. ORB improves upon this by ranking the FAST keypoints using a Harris corner measure so it is able to filter out any unstable points and keep the corners. This helps us to ensure that our reference stop sign and test images are using consistent high quality features. oFAST incorporates rotation by computing the intensity centroid of the image area that is nearby the keypoints.

It uses the image moment:

$$m_{pq} = \sum_x \sum_y x^p y^q I(x, y) \tag{1}$$

to compute the centroid:

$$C = \left(\frac{m_{10}}{m_{00}}, \frac{m_{01}}{m_{00}} \right) \tag{2}$$

and orientation angle:

$$\theta = \text{atan2}(m_{01}, m_{10}) \tag{3}$$

It then stores the theta of the rotation as a descriptor.

C. BRIEF and rBRIEF: Creating Descriptors

BRIEF creates a 256 bit descriptor by sampling intensity pairs inside small patches around the keypoints. rBRIEF rotates the BRIEF sampling pattern using the angle theta from oFAST using a rotation matrix to align the sampling pairs with the orientation keypoint.

$$S_\theta = R_\theta S \tag{4}$$

This allows the descriptors from the test stop signs to match consistently with the reference template no matter the rotation.

D. RANSAC: Geometric Verification

After we have extracted ORB descriptors from both images we are comparing we perform brute form matching using hamming distance. These raw matches contain many outliers from background clutter and edges present in urban environments. We use RANSAC for geometric consistency and estimate a homography between the reference and test image. RANSAC randomly selects a set of four matched pairs and computes their homography and evaluates how many inliers this transform produces. It then repeats the sampling and only keeps the homography with the highest inlier count. This step is important to our system since only a stop sign should consistently map to the reference image. We then classify the image as containing a stop sign or not if the number of inliers determined by RANSAC reaches the set threshold. Experimentally we chose 9 as the minimum number of inliers.

IV. Dataset Creation

A key part of this experiment is the creation of the dataset. We aimed to capture images representing typical urban intersections around Boston. These can be cluttered and contain a lot of distractions, such as murals and other road signs in the background. We obtained 5 categories of images and 3 categories of videos to test the accuracy of ORB on different angles and distances. We found nine stop signs located around Northeastern University set in different environments, with their locations given in the map below. With each stop sign, we obtained “close up”, “far away”, and “angled” pictures. These were taken on an iPhone on standard camera settings at approximately 3:00 pm on an overcast day. The “close-up” picture group was taken level with the stop sign, approximately 3-5 feet away. The “far away” images were also level with the sign, but close to 15 to 20 feet away. The angled group has pictures taken from various downward angles around the sign, corresponding to where a car would be, also around 3-5 feet away. We also obtained 10 street images that would be observed by a car with no road signs visible. The last group included 8 images taken from 3 to 5 feet away of road signs that were not stop signs in order to test and prove success for the false positive case. For the video portion, we obtained videos from the angle of a typical sedan driving towards a stop sign at around 10 miles per hour. Three of these included stop signs, one with a red “do not enter” sign, and one with no signs at all to test for all cases.

Fig. 1: Video Searching Frame Example

V. ORB + RANSAC with Modifications

Our algorithm begins by resizing and reformatting images for processing. We found that the smartphone camera used for collection produced computationally complex images with very high resolution. We had to first convert the images to a processable .jpg format. Then, to further combat the issue, we resized the images and added a Gaussian blur to simulate motion blur. Similarly, we resized and reduced the video resolution for easier computation in order to process them frame by frame, like the image example. In the video case, we also added checks for the aspect ratio and relative feature size to account for feature movement throughout the video. The algorithm begins with using ORB for initial feature detection in the reference and test images for both picture and video cases. Next, Brute Force matching (BF Matching) is used along with the Hamming distance to match features between the two images. Then, RANSAC is applied to find the relationship and translation between an example feature in both images. Once the homography matrix is generated, it defines the geometric relation between the two images and can be used to filter inliers and outliers. Based on our inlier threshold, the algorithm determines if the feature (stop sign) is found in the test picture. Our threshold was set at 9, which seemed to provide the most accurate results across all cases.

VI. Detection System Overview

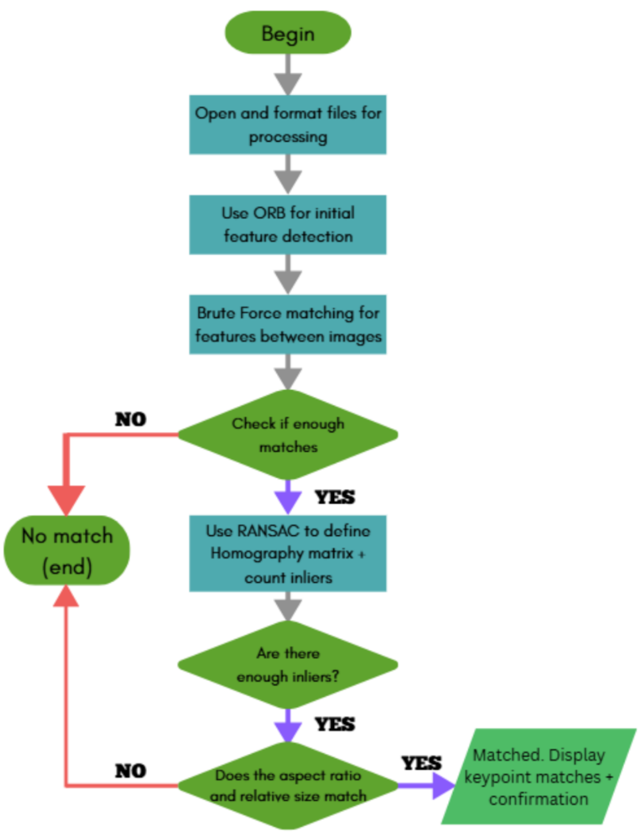

Fig. 2: System Block Diagram

VII. Results

A. Image Detection

Upon executing the algorithm described at the optimal threshold values we determined our results are displayed in the table. The corresponding images with the stop signs and their detection results are displayed in images below.

TABLE I: Image Detection Accuracy

Image Category	Accuracy
Close Up	85%
Angled Stop Sign	55%
Far Away Stop Sign	0%
No Stop Sign	100%
Other Road Signs	100%

Fig. 3: Close Up Stop Sign: 85% Accuracy

Fig. 4: Angled Stop Sign: 55% Accuracy

Fig. 5: Far Away Stop Sign: 0% Accuracy

Fig. 6: No Stop Signs: 100% Accuracy

We can see that the algorithm performed well on close up images with a score of 85%and was less accurate for images taken at an angle with a score of 55% accuracy. It performed the worst on the far away section at 0% detection rate for images taken 20 feet away. It performed very well and did not produce any false positives for detection in images with no stop signs or other road signs with a 100% accuracy score for both sections. The original ORB paper evaluates feature matching on controlled datasets; however our evaluation focuses specifically on stop-sign detection in real outdoor environments. Therefore, our performance metrics cannot be directly compared to the original benchmarks

B. Video Detection Results

Again, the video case operated with a much smaller dataset of six total videos, four with a stop sign, one containing a "do not enter" sign, and one with no signs at all. The results are displayed in the table below. This was done from an angle akin

TABLE II: Video Detection Accuracy

Video Category	Accuracy
With Stop Sign	75%
With Other Sign	100%
Without Signs	100%

Fig. 7: Other Road Signs: 100% Accuracy

to the driver's seat of a modern vehicle, while at a relatively low speed compared to regular road traffic. A unique issue that arose in this case was the algorithm misdetecting small red features as a stop sign during frames when the real stop sign was too small or out of the image. We added checks for relative feature size to account for this, ensuring that the detected feature remains at a size close to the reference image and consistent throughout the video. We also had to take aspect ratio into consideration, as the video would change perspective angles as the camera moves. The test results can be seen in the table below. A successful result was categorized as a case where the algorithm can detect the sign's existence as the camera begins approaching the sign and can remain at that decision until the sign is out of frame.

While in future cases we would like to test this over a larger dataset, we were able to see that the algorithm was relatively successful at correctly determining the result. It was also most successful in the intermediate distance range, while the sign is neither too close nor too far. The sign was also found even through sharp angle changes due to movement. However, we found that we could not move too fast as the motion blur affected the algorithm's ability to distinguish features. A gimbal or some sort of stabilizer function, along with a camera with higher frame rate capture capability would help in this case.

VIII. Conclusion

In this work we evaluate the performance of ORB for feature detection combined with RANSAC based geometric verification for stop sign detection in real urban environments around Boston. Our results show that the algorithm performs reliably on images taken close up and front facing achieving 85% accuracy. ORB was also able to detect angled stop signs with an accuracy of 55%. The system struggled significantly with

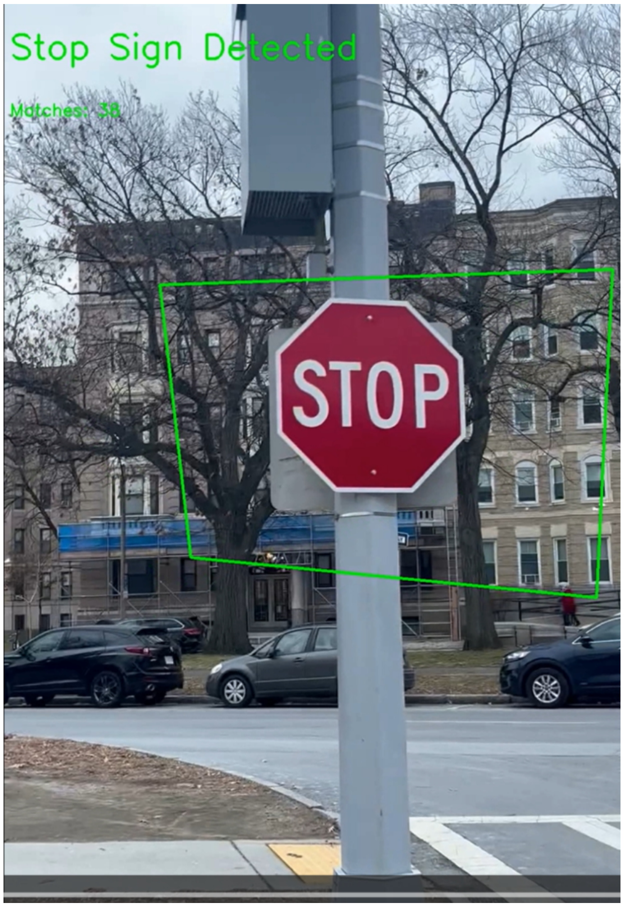

Fig. 8: Video Detection Frame Example

Fig. 9: Video Searching Frame Example

scale variation and could not detect any stop signs from 20 feet away. Our results align with known limitations of ORB which is very reliant on the density and quality of detected keypoints.

Limitations and Challenges

We encountered several practical challenges during implementation including sensitivity to image resolution. To solve this challenge we had to preprocess the image and resize it as well as adding gaussian blurring to mimic realistic conditions. We also needed to manually set the inlier threshold which introduced variability in results across the different images. Despite these limitations the algorithm did not produce any false positives on other road signs or images containing no stop signs. This shows that RANSAC was very effective at filtering out geometrically inconsistent matches.

Future Work

Future work may explore incorporating new preprocessing image techniques to improve accuracy in low visibility situations. Expanding the dataset to include more versatile urban scenarios taken at different times of day and in different weather would further strengthen our evaluation. Overall our results indicate that ORB and RANSAC provide a fast approach for stop sign detection, the performance is very dependent on the image quality and scale suggesting a need for more advanced detection algorithms.

References

[1] E. Rublee, V. Rabaud, K. Konolige, and G. Bradski, “ORB: An efficient alternative to SIFT or SURF,” in Proc. IEEE Int. Conf. Comput. Vis. (ICCV), Barcelona, Spain, Nov. 2011, pp. 2564–2571.

[2] GeeksforGeeks, “Step-by-step guide to using RANSAC in OpenCV using Python,” https://www.geeksforgeeks.org/python/step-by-step-guide-to-using-ransac-in-opencv-using-python/ (Accessed: Dec. 1, 2025).